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**Original Article**

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# Original Article

# Experimental evidence of mate choice as the driving mechanism behind negative assortative mating for shell colour in a marine snail

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## ABSTRACT

How colour polymorphisms are maintained in natural populations constitutes a key subject of study for evolutionary ecologists. One of the mechanisms that has been put forward to explain this phenomenon is negative frequency-dependent selection, which can be caused by different mechanisms. Among these mechanisms is mate choice, or the selection as a potential mate of the rare or dissimilar colour morph by the choosy sex of a given species. In the context of colour polymorphism, mate choice predicts the occurrence of a negative assortative mating pattern, whereby different colour morphs mate more often than expected by chance alone. However, negative assortative mating can also result from other mechanisms, making it, at best, a fallible indicator of a mate choice-driven negative frequency-dependent selection. For this reason, experimental evidence explicitly connecting such a mating pattern to mate choice is necessary to claim that a particular colour polymorphism is being maintained by negative frequency-dependent selection. Here we present experimental evidence backing this relationship for the colour polymorphic marine gastropod *Littorina saxatilis* from the Ría de Vigo (NW Iberian Peninsula), although how specifically these organisms choose their mates remains unclear. This calls for further experimental efforts to clarify this issue and its relationship to the maintenance of colour polymorphism in *L. saxatilis*.

**Keywords:** colour polymorphism; *Littorina saxatilis*; mate choice; negative assortative mating; negative frequency-dependent selection; shell colour

## INTRODUCTION

The evolution and maintenance of colour polymorphism is a classic topic of study in evolutionary ecology (Ford 1945, 1979, Bond 2017, Svensson 2017). Most major animal taxa contain at least some species that show intrapopulation colour diversity (Gray and McKinnon 2007, White and Kemp 2016), which prompts the question of how this condition is preserved in spite of the homogenizing forces of natural selection and genetic drift (Lewontin 1974, Nielsen 2005). The presence of colour polymorphism has been linked to several important evolutionary processes, such as speciation and diversification (Gray and McKinnon 2007, Hugall and Stuart-Fox 2012, Kusche *et al.* 2015, Brock *et al.* 2022), or population persistence (Forsman *et al.* 2008). Coloration in animals may have three different (and nonexclusive) fitness-related functions: communication (with either conspecifics or predators, or both), crypsis, and thermoregulation (Endler 1978). These functions, in conjunction with microhabitat or spatial heterogeneity, can lead to the evolution of colour polymorphisms through divergent selection (Gefaell *et al.*, 2023a). For instance, there is evidence that the shell colour polymorphism shown by the populations of the marine snail *Littorina obtusata* from the Gulf of Maine (which live under different thermal regimes) is maintained by selection on thermal stress resistance (Phifer-Rixey *et al.* 2008).

Another possibility regarding the evolution and maintenance of colour polymorphism—or for that matter, of any other kind of genetic polymorphism—is negative frequency-dependent selection, in which the rarer colours (or their corresponding alleles) are protected from extinction because of their greater relative fitness (Ayala and Campbell 1974, Futuyma and Kirkpatrick 2017,

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Brisson 2018, Sætre and Ravinet 2019). Negative frequency-dependent selection, in turn, can take place through different ecological and/or behavioural mechanisms, such as resource competition, apostatic selection, or mate choice of rare or dissimilar individuals (Kokko *et al.* 2007, Chouteau *et al.* 2017, Brisson 2018, Estévez *et al.* 2020). The role of apostatic selection as the driving force of negative frequency-dependent selection for colour variation has been widely studied over the last 60 years, with the classic model genus *Cepaea* standing out in this regard (e.g. Clarke 1962, Tucker 1991, Cook 2017). In such a system, apparently individuals with the least frequent colour patterns go unnoticed by at least some of their predators (particularly the song thrush, *Turdus philomenos*), which seem to prey by forming search images of their most common prey (Tucker 1991; but see Cook 2017). The mechanism of mate choice has also been investigated, and some putative examples have been described in nature (Hughes *et al.* 2013, Wellenreuther *et al.* 2014, Estévez *et al.* 2020, Potter *et al.* 2023). According to the mate choice hypothesis, the individuals of the choosy sex of a given species would show a consistent behavioural preference either for those conspecifics of the opposite sex that bear the least frequent colours and/or alleles, or for those that are simply different (i.e. dissimilar) from themselves, depending on whether the choice is for rarity or dissimilarity. In the mid- to long term, this preference would lead to an (inter)sexual selective advantage of the rare and/or dissimilar colours and, thereby, to their protection from extinction (Wellenreuther *et al.* 2014, Estévez *et al.* 2020).

An indicator of mate choice-driven negative frequency-dependent selection in a population can be negative assortative mating (also known as disassortative mating), in which the frequency of matings between individuals with different phenotypes or genotypes is higher than that expected by chance alone (Follett *et al.* 2007, Jiang *et al.* 2013). Assortative mating and its relationship to colour polymorphism have also been studied in several taxa (Calderón *et al.* 2010, Pryke 2010, Pérez i de Lanuza *et al.* 2013, Fargevieille *et al.* 2017). Additionally, support for the link between negative frequency-dependent selection and negative assortative mating has been obtained at both the empirical and the theoretical levels (Pusey and Wolf 1996, Takahashi and Hori 2008, Field and Barrett 2012, Holman *et al.* 2013, Hedrick *et al.* 2016, 2018, Estévez *et al.* 2020), which means that the presence of a negative assortative pattern in a population could be counted as evidence, if only weak, of negative frequency-dependent selection. However, it is also well known that (negative) assortative mating can result from other population processes or mechanisms (Hendry and Day 2005, Taborsky *et al.* 2009, Snowberg and Bolnick 2012, Wellenreuther *et al.* 2014, Kopp *et al.* 2018). For this reason, to argue for a mate choice-driven negative frequency-dependent selection, it is not enough to corroborate the existence of a negative assortative mating pattern for colour, but also to provide direct and independent evidence that this mating pattern is brought about by an active choice of the rare or dissimilar colour morph(s) by the choosy sex. Taken together, both pieces of evidence (negative assortative mating and mate choice) would constitute a more robust suggestion of negative frequency-dependent selection driving the maintenance of a colour polymorphism.

*Littorina saxatilis* is a marine gastropod that lives in the supralittoral zone of intertidal ecosystems across the northern American and European coasts of the Atlantic Ocean (Reid 1996, Rolán-Alvarez *et al.* 2015a). This species has been widely used as a model organism for the study of evolutionary processes, as its ovoviviparous mode of reproduction curbs its dispersal capability, leading to the isolation of its populations and to the evolution of (sometimes recurrent) local adaptations (Rolán-Alvarez 2007, Galindo and Grahame 2014, Rolán-Alvarez *et al.* 2015a, Johannesson 2016). *Littorina saxatilis*, like other species of the genus *Littorina*, also shows a remarkable and genetically based shell colour polymorphism, with shell colours ranging from whitish cream to black, to which several patterns can be added (lines, tessellations, and bands) (Reid 1996, Rolán-Alvarez *et al.* 2015a, Johannesson and Butlin 2017, Gefaell *et al.* 2023a). A particularly remarkable area in terms of the colour polymorphism shown by *L. saxatilis* is the Ría de Vigo, a coastal inlet in the NW Iberian Peninsula that shows both intrapopulation and clinal colour diversity in the populations of this species (Sacchi 1979, Reid 1996, Gefaell *et al.* 2023a, b). More specifically, the populations of *L. saxatilis* living in the wave-sheltered localities of this Ría show a recurrent intrapopulation shell colour polymorphism, with snails coexisting that can be divided into whitish or greyish-hazel, yellow, and brown colour morphs (Gefaell *et al.* 2021, 2023; see Fig. 1A, B and Supporting Information, Fig. S1A). *Littorina saxatilis* and other species of this genus constitute an excellent system for studying sexual selection-related processes, as the matings of this species are easily detectable in the field (during low tide) and show a relatively high prevalence in most populations (Rolán-Alvarez *et al.* 2012, 2015a, b). In *Littorina*, males appear to be more active and selective in the search for potential mates, and females apparently do not exert too much choosiness in their mating behaviour, showing multiple paternity (Panova *et al.* 2010, Saltin *et al.* 2013, Ng *et al.* 2019).

In a previous study, evidence was obtained using a validated protocol (e.g. Rolán-Alvarez *et al.* 2015b, Ng *et al.* 2016, Estévez *et al.* 2018) that *L. saxatilis* from the wave-sheltered localities of the Ría de Vigo, as well as their sister species *Littorina fabalis* and *L. obtusata* from other ecologically similar populations, show a consistent negative assortative mating pattern in the wild, with lighter coloured individuals mating significantly more often with darker ones than expected by chance (Gefaell *et al.* 2021). This result was interpreted as suggesting that (i) an ancestral behavioural mechanism of mate choice for colour dissimilarity could be driving the negative assortative mating pattern found in these *Neritrema* species (the subgenus comprising *L. saxatilis*, *L. fabalis*, and *L. obtusata*, as well as other related species); and (ii) that this would bring about the maintenance of their shell colour polymorphisms through negative frequency-dependent selection. However, in accordance with the recommendation outlined above, these conclusions would not be warranted unless we could find independent evidence of mate choice in at least one of these species. This evidence has been found in wild populations of *L. fabalis* (Estévez *et al.* 2020), but no further investigations have been carried out so far in *L. saxatilis* or in a laboratory setting.

In the present study, we employ an experimental laboratory paradigm consisting of mating balls to establish whether the negative assortative mating pattern detected in the wild populations of *L. saxatilis* from wave-sheltered localities can be attributed to a mate

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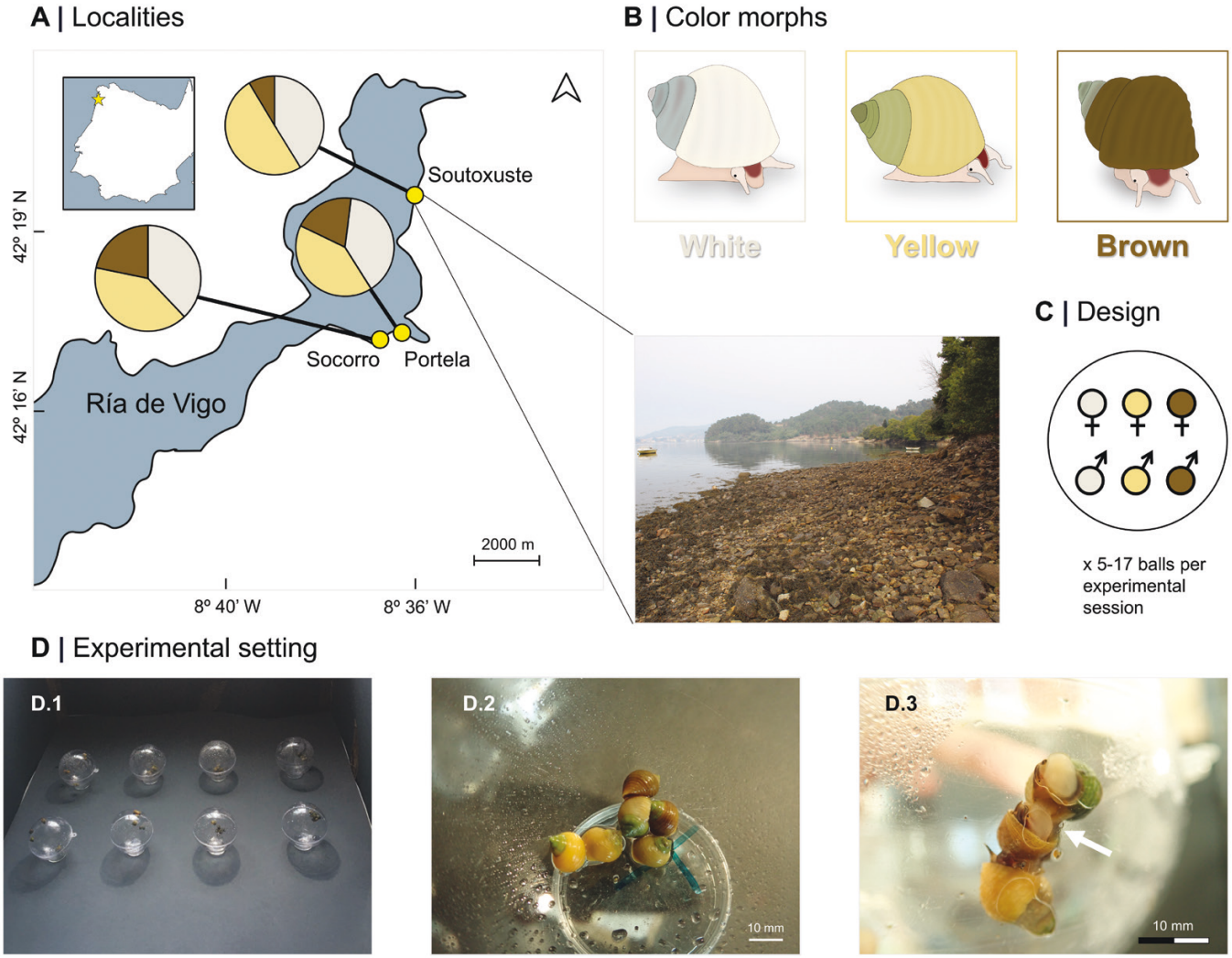



**Figure 1.** A, location of the localities studied within the Ría de Vigo with their corresponding frequencies of colour morphs (taken from Supporting Information, Table S1), and a photograph of Soutoxuste as representative of the three localities (map taken and modified from https://www.d-maps.com). B, drawings of the three colour morphs involved in this study (white, yellow, and brown). C, our adaptation of the experimental paradigm employed, which consists of a series of Christmas (mating) balls in which individuals from each sex (male, female) and colour (white, yellow, and brown) are introduced to record their mating behaviours. D, experimental setting devised using the mating balls. D.1 shows the balls in an experimental session, inside a cardboard box that had been covered with a black paper box and illuminated with artificial light. D.2 shows the six snails (two sexes from each of the three colours) inside a mating ball at the beginning of an experimental session. The yellow ones are on the left, followed by the whites and the browns. D.3 shows an instance of mating as seen from below. The penis of the male (white arrow) is entering the aperture of a female (diagnosed by the absence of a penis), which according to the operational definition outlined in the text counts as a mating.

choice behaviour for colour dissimilarity. If so, that would constitute a further piece of evidence in favour of the negative frequency-dependent selection scenario for the maintenance of shell colour polymorphism in these localities. In accordance with the natural history of this species, as well as our previous results on this matter (Gefaell *et al.* 2021), our hypothesis for this experiment was that males of *L. saxatilis* would show a consistent behavioural preference for females (i.e. mate choice) with a different shell colour that would result in a negative assortative mating pattern.

## MATERIALS AND METHODS

### Localities studied and sampling procedure

The snails used in the laboratory experiment were collected during the summer (end of June and the whole of July) of two consecutive years (2021 and 2022) in three different localities of San Simón Bay, in the inner zone of the Ría de Vigo and close to the mouth of the Verdugo River: Soutoxuste, Portela, and Socorro (see Fig. 1A). The first two localities (Soutoxuste and Portela) were also sampled for a previous study in 2020 (Gefaell *et al.* 2021), and for the present one both were sampled in 2021 and 2022, while the third locality (Socorro) was sampled only in 2022. Soutoxuste, Portela, and Socorro constitute wave-sheltered spots around 100 m long and 15 m wide formed by small alkaline granitoid rocks no larger than 20–30 cm in size (see the photo in Fig. 1A). Although these localities do not experience any swell, they are fully covered at high tide (especially during spring tides), so the sampling sessions always took place at low tide. Soutoxuste is separated from Portela by 7380 m of coast and from Socorro by 7960 m, which means that Portela

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and Socorro are around 580 m apart (see Table 1 for GPS coordinates). From the point of view of *L. saxatilis*, however, these distances are large enough to consider the populations from these localities as at least partially genetically isolated (Rolán-Alvarez *et al.* 2004, Tirado *et al.* 2016). All three localities were selected based on their high density of *L. saxatilis*, while the season for conducting the experiment (i.e. summer) was chosen because of the higher mating propensity of this species at this time of year (in part because the weather is better; E. Rolán-Alvarez, personal observation). The populations of *L. saxatilis* from Soutoxuste, Portela, and Socorro, as with the rest of populations from other wave-sheltered localities in the Ría de Vigo, are composed of three main colour morphs bearing different background colorations and that are present at relatively stable frequencies (see pie charts in Fig. 1A and Supporting Information, Table S1): whitish or greyish-hazel (hereafter, white), yellow, and brown (in the last case, sometimes with thin slight lines along the grooves of the shell) (see Fig. 1B and Fig. S1A).

These three colour morphs were used for our experiment. In other publications studying the colour polymorphism of *L. saxatilis* from the Ría de Vigo, a different colour classification system than the one employed for this study has been used (Sacchi 1979, Gefaell *et al.* 2023b). Under this system, most of the snails labelled 'white' in the present study would fall under the category '*fulva*', which would also cover the lightest yellow and brown individuals. For the present study, however, we stuck to the colour classification system employed by Gefaell *et al.* (2021) to have a common basis from which to make meaningful comparisons between both studies (something that philosophers of science call 'commensurability'; Kuhn 1982). Although occasional assignation errors may have occurred when classifying the snails for this study, major or systematic errors were avoided by picking up only the clearest and distinctive white, yellow, and brown-shelled individuals (see Supporting Information, Fig. S1A), and discarding those showing less vivid or more intermediate colorations between the three selected classes. Snails with microalgae growing on them, or with worn and/or damaged shells were also excluded.

Our experiment was divided into discrete experimental sessions, so the samples for each of these sessions were always collected in the field the day before. The snails were released to the locality where they were collected either after the last experimental session of the season (in 2021), or immediately after finishing their corresponding experimental session (usually 2 days after being collected; in 2022). This last release strategy drastically reduced the stress and mortality rate of the snails used. To avoid resampling (and consequently, retesting) previously used snails, in each sampling session at a given locality we only picked individuals from a small spatial range (4–5 $m^2$). The selection of snails was also carried out taking into account shell size, since positive assortative mating for this trait has been previously described in *L. saxatilis* and other species of the family Littorinidae (Saltin *et al.* 2013, Fernández-Meirama *et al.* 2017, Lau *et al.* 2021). For this reason, only snails converging around the mean shell size of the population (e.g. shell length of ~10 mm) were collected. This resulted in relatively homogeneous samples per locality. For transportation from the field to laboratory (and vice versa), all snails collected were introduced into separate plastic boxes by colour (white, yellow, and brown) that were previously arranged so that they simulate the natural habitat of the snails, with small rocks and brown algae (mainly *Ascophyllum nudosum*, *Pelvetia canaliculata*, and *Fucus* sp.) taken from the localities (see Supporting Information, Fig. S1B). Since *L. saxatilis* is not a protected species and is present in large numbers throughout the Ría de Vigo, no special permissions were needed to conduct this experiment. However, we tried to adhere to the ASAB guidelines (ASAB Ethical Committee/ABS Animal Care Committee 2023) during all steps of the process.

### Sex determination and laboratory storage

The live snails were taken to the laboratory facilities of the Toralla Marine Science Station (ECIMAT), ~30 min from the localities by car (35–40 km), avoiding aggressive movements or sounds, and maintaining a moderate temperature (~18°C) by using the car's air conditioning. Once at ECIMAT, the individuals were sexed and stored in separate water tanks (by colour and sex) recreating a natural environment (again, with small rocks and brown algae from their localities) and filled with constant running water from the ocean. Sex determination was carried out by visual inspection of the snails with the aid of a stereo microscope (see Supporting Information, Fig. S1C). Males were diagnosed by the presence of a large protuberance (the penis) in the right dorsolateral portion of their bodies, while females were diagnosed by the absence of such a trait. The presence of immature individuals was ruled out since the range of shell sizes of the snails used in this experiment corresponds to that of mature individuals for this species. Once the sex determination process ended, the snails were stored in their water tanks for between

**Table 1.** Summary of the sample characteristics of each locality studied (with their corresponding GPS coordinates).

| | GPS coordinates | Sample characteristics | | | |
|---|---|---|---|---|---|
| | | $n_1/n_2$ | $N_1/N_2$ | Male size | Female size |
| Soutoxuste | 42°19′21.19″N, 8°36′55.73″W | 33/13 | 198/78 | 9.2 ± 0.53 | 9.5 ± 0.56 |
| Portela | 42°17′17.87″N, 8°37′22.16″W | 57/48 | 342/288 | 10.3 ± 0.88 | 10.6 ± 0.91 |
| Socorro | 42°17′16.16″N, 8°37′37.53″W | 38/38 | 228/228 | 10.7 ± 0.95 | 11.1 ± 0.90 |

$n_1$ is the number of mating balls per locality used in this experiment for which there was at least one mating event (total $N = 128$), while $n_2$ is the subsample of those balls for whose snails the shell size was measured (most of them corresponding to the 2022 experimental sessions; total $N = 99$). $N_1$ is the number of individuals per population that participated in laboratory experiments, and $N_2$ is the subsample of those in which shell size was recorded (note that $N_1 = n_1 \times 6$ and $N_2 = n_2 \times 6$; as there were six individuals per mating ball or experimental unit; see text for more details). Mean ± standard deviation of male and female size (mm) for that subsample of snails in which shell length was measured ($N_2$) is also shown, also grouped by locality.

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4 and 31 h for acclimation, with a natural day/night light cycle until their corresponding experimental sessions began (see Fig. S1D).

### Mating ball experiment design

For the mate choice experiment, we followed an experimental paradigm similar to that of Lau *et al.* (2021) to measure positive assortative mating for shell size in *Echinolittorina malaccana*. In that experiment, male and female individuals of different sizes were placed in transparent plastic Christmas balls for a predefined time period and the matings between them were recorded. This experimental paradigm allows male snails to chase females by following their mucus trails (as is usual in the family Littorinidae; Erlandsson and Kostylev 1995, Ng *et al.* 2011, 2019, Saltin *et al.* 2013), so that it is possible to link the resulting mating pattern to a behavioural mechanism of (male) mate choice.

In our adaptation of the experimental paradigm of Lau *et al.* (2021), snails of each colour and sex ($N = 6$ individuals, 3 per colour × 2 per sex) were introduced into a mating ball (diameter = 10 cm; see Fig. 1C, D, as well as Supporting Information, Fig. S1E), and their mating behaviours during the next 3 h were recorded following a scan sampling protocol (Bateson and Martin 2021) with behavioural recordings every 3 min, as the available data suggest that the most common matings in this species usually last around 4 min (Perini *et al.* 2022). This 'multiple-choice' design, in which three different males (white, yellow, and brown) are placed together in a single ball with three different females of equivalent colours, was favoured over 'single-choice' (one male and one female of each possible colour combinations) or 'male-choice' designs (one male of a given colour and the corresponding three different coloured females), as the former scenario is more biologically realistic (*L. saxatilis* tends to live and mate at high densities of both males and females). Furthermore, in their experiment, Lau *et al.* (2021) revealed that only the multiple-choice paradigm (but not the single- or male-choice paradigms) resembled the mating patterns observed in the wild in *Echinolittorina malaccana*, which shares both its taxonomic family (Littorinidae) and its ecological niche with *L. saxatilis*. For our experiment, the working hypothesis was that, if a male mate choice behaviour for dissimilar females does exist (as revealed in the field; Gefaell *et al.* 2021), then the mating pattern observed in the laboratory should also be disassortative (i.e. negative assortative).

Throughout the whole experiment, we employed a total of 192 mating balls. These were divided into different 18 experimental sessions of 5–17 mating balls each, depending on the number of snails collected and sexed each sampling day. Since males are the choosy sex, females were always introduced into the mating balls first. As for the colour morphs (white, yellow, brown) of both sexes, their order of introduction was randomized. The size of all snails introduced into each ball was homogenized to account for positive assortative mating for size, selecting those with a similar shell length under visual inspection. However, taking into account that, despite the overall positive assortative mating for size, previous results also have shown a preference of male littorinid snails for slightly larger females (Saltin *et al.* 2013, Ng *et al.* 2019, Lau *et al.* 2021, Clancey *et al.* 2022), we also tried to choose females that, despite their overall size similarity to the males, were slightly larger than these to control such an effect in our experiment. In addition to the size homogenization, for most of the 2022 sessions the shell length of all snails used was also recorded to the nearest 0.1 mm (from apex to the tip of the last whorl) using a vernier caliper to formally rule out the effect of size when explaining the resulting mating pattern. Before the start of each experimental session, all balls were sprayed with ocean water (sometimes containing pulverized brown algae) to recreate the natural environment of the snails. This spraying was repeated once every hour during the 3-h experimental session to keep the balls moist. To minimize the influence of extraneous environmental variables, all balls employed in each experimental session were placed on a cardboard box covered with a black paper board and illuminated with a constant artificial light (Fig. 1D.1).

We determined that a male had chosen a mate when we observed a copulation. We defined a mating when two requirements were met: (i) when inspecting the snails from above, one snail (usually the male) was placed with the aperture of the shell facing up such that it overlapped with the aperture of a second snail (usually the female), which was located face down, with both snails forming a V-like pattern (Gefaell *et al.* 2021); and (ii) when close inspection from below revealed that the penis of the male (faced up) was inside the aperture of the female (faced down) (see Fig. 1D.3). Behavioural recordings were carried out by a single experimented observer (Juan Gefaell) to avoid intersubject reliability problems. No blinding procedure was implemented, as it was strictly necessary to see the colours involved in the matings. Although we recorded all matings that took place during the experimental sessions, only the first one in each ball was initially considered for the statistical analyses, as the subsequent matings cannot be considered independent observations (see further details in the section below). No simultaneous matings occurred when it came to the first mating, so this was always defined as the one that occurred first at a given ball. Once the experimental session was over, the snails were removed from the mating balls and stored in separate tanks, prior to being carried back to the field (see Supporting Information, Fig. S1F). Before a new experimental session, all mating balls used were cleaned with fresh water to remove the mucus trails from the previous session.

### Statistical analyses

For data analysis, the mating ball was treated as the experimental unit. In each ball, there could be a total of $3^2 = 9$ possible mating types, corresponding to the combinations of each colour (white, yellow, brown) and sex (male, female); these, in turn, can be divided into 'homotypic' (with both sexes showing the same colour) or 'heterotypic' (with different colour) matings. Three possible outcomes were possible for each experimental unit (i.e. mating ball): (1) no matings during the 3 h; (2) just one mating; and (3) more than one mating. All the experimental units in which outcome (1) occurred were discarded, keeping only those that met outcomes (2) and (3). Male–male matings, relatively common in this species (Rolán-Alvarez *et al.* 1999), were also excluded to avoid possible confounding factors. As previously stated, for balls with outcome (3) only the first mating was incorporated into our initial analyses, as it is the only one that meets the assumption of the independence of data (i.e. the ensuing matings

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could be somehow influenced by the previous ones, especially when the same male individual mated more than once per experimental session; Bateson and Martin 2021). The frequencies of the nine possible mating types can be expressed through a nine-cell mating table; a homogeneity G likelihood test (Sokal and Rohlf 1995) was used to compare the frequencies of the different mating types obtained in the first matings of outcomes (2) and (3) to those obtained in the subsequent matings of outcome (3). When the frequencies in the former test were clearly homogeneous for a given locality ($P > .25$), these were nonetheless grouped together to increase the sample size.

Of the total of 192 mating balls, 128 (67%) met criteria (2) or (3) outlined above (see Table 1). These were then divided into two separate groups based on whether the size of the snails in each of these balls was measured or not (mostly corresponding to the 2022 and 2021 series, respectively; see Table 1). This allowed for two independent analyses of matings: a logistic regression analysis in which size was included as an explanatory factor for the type of mating (homotypic vs. heterotypic; see paragraph below), and a second analysis in which the variable size was ignored and the mating pattern was studied using the $I_{PSI}$ estimator (Pérez-Figueroa *et al.* 2005). All analyses were carried out separately for each locality (Soutoxuste, Portela, Socorro).

Despite the fact that the size of the snails was homogenized both when collecting the snails in the field and during the setup of the mating balls for the experimental sessions, it could still be conceivable that this parameter could exert some influence over the observed mating pattern. Therefore, a logistic regression analysis was carried out to explicitly rule out this possibility. For this analysis, only the subsample for which shell size was measured was used. The first mating observed in each ball was employed as the dependent variable, and coded either as a homotypic mating ('1') or as heterotypic ('0'). The size of the snails involved in the matings, as well as their interaction, were employed as predictor variables in the logistic regression model.

The $I_{PSI}$ analysis was performed using the JMATING v.1.0 software (Carvajal-Rodriguez and Rolan-Alvarez 2006) to examine the overall direction of the pattern of matings recorded. The $I_{PSI}$ estimator is an index of assortative mating that ranges between −1 (complete negative assortative mating) and 1 (complete positive assortative mating), with 0 being complete random mating. The $I_{PSI}$ index has proven to be a robust and reliable estimator of assortative mating for qualitative traits under different sampling assumptions (Pérez-Figueroa *et al.* 2005, Rolán-Alvarez *et al.* 2015b, Estévez *et al.* 2020). In the present experiment, the $I_{PSI}$ index was calculated from a table summarizing all actual matings between colour morphs in each locality (normally the first ones except when the homogeneity tests allowed the inclusion of the subsequent matings, which happened for Soutoxuste and Portela, but not Socorro; see Supporting Information, Table S2) under two different scenarios: each colour and sex plotted separately against each other, and every single colour plotted against the rest of the colours grouped together (e.g. white vs. yellow and brown grouped together). Statistical significance for $I_{PSI}$ values was calculated through bootstrapping, resampling 10 000 times the frequencies of observed mating combinations for the different colour morphs. The existence of a negative assortative mating pattern was studied through a maximum likelihood G test, comparing the frequency of observed matings of each kind with their corresponding expected values in a scenario of random mating for colour. The absence of 'size' as a potential variable contributing to the observed pattern in this test was countervailed by the previously described logistic regression analysis. All statistical analyses were carried out in Excel and SPSS v.24.0 (IBM Corp., USA).

### Mate choice estimation through model selection

The previous analyses were complemented with an estimation of mate choice made by connecting the empirical data with explanatory mathematical models. This was done using the InfoMating software v.0.4 for model selection and parameter estimation (Carvajal-Rodríguez 2020), using the same samples of matings as for the $I_{PSI}$ analysis. The methodology behind this estimation is that InfoMating works with several alternative models for the parameters of mate choice and/or mating propensity (in our case it considers up to 23 alternative models). These mechanisms (mate choice and mating propensity) have traditionally been considered the main explanations for selective mating under controlled laboratory conditions (O'Donald 1980, Carvajal-Rodríguez 2018, Lau *et al.* 2021). InfoMating calculates the maximum likelihood values of the parameters of each model. These values are then used to rank each model according to its fit to the data following the Akaike information criterion corrected for low sample size (AICc; Carvajal-Rodríguez 2018, 2020). The purpose of this analysis is to detect mate choice coefficients ($C$) in the nine-cell mating table of different mating types—i.e. values that are distinct from the random mating expectation coefficient (corresponding with a value of 1). If $C < 1$ values are shown by the model in the diagonal of homotypic mating combinations (those with identical male and female colours), then that would represent a negative assortative mating pattern (for formal details, see Carvajal-Rodríguez 2018, 2020). Different models propose different $C$ coefficients ($C_1, \ldots, C_n$) to account for the various mating types. A model that presents a unique $C$ for each of the nine possible mating types in the nine-cell mating table ($C_{1-9}$) would represent the same coefficients as the PTI (pair total isolation) statistic, as they stand for the maximum likelihood model with the same number of parameters as cells in the mating table (Carvajal-Rodríguez 2020). The PTI also constitutes the deviation of observed mating frequencies in each of the nine possible mating types, with respect to the random mating expectation based on initial colour frequencies (Rolán-Alvarez and Caballero 2000).

## RESULTS

The logistic regressions for each locality (Soutoxuste, Portela, and Socorro) are summarized in Table 2. Male and female shell sizes (as well as their interactions) were not significant in any of the logistic regression models, as shown by their low $R^2$ values and their lack of significant contribution to the explanation of the dependent variable. This points to an absence of effect of the variable 'size' on the overall mating pattern and supports our procedural effort to homogenize the shell size of the snails used in the experiment. For this reason, the subsequent analyses ignored size as an explanatory factor.

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**Table 2.** Logistic regression of snail size parameters (male size, female size, and their interaction) as potential predictors of mating type ('Homotypic mating or not', with same colour–homotypic–mating = 1 and different colour–heterotypic–mating = 0; see the text for a detailed explanation).

| Dependent | Locality | $n_2$ | $R^2$ | Variables studied | *B* | SD | *P* |
|---|---|---|---|---|---|---|---|
| Homotypic mating or not | Soutoxuste | 13 | 0.205 | Constant | 440.3 | 726.2 | .544 |
| | | | | Male size | −45.3 | 75.6 | .56 |
| | | | | Female size | −49.3 | 75.6 | .514 |
| | | | | Male size × Female size | 5.1 | 8 | .528 |
| Homotypic mating or not | Portela | 48 | 0.04 | Constant | −1.7 | 61.3 | .978 |
| | | | | Male size | 0.5 | 6.1 | .929 |
| | | | | Female size | 0.3 | 5.9 | .965 |
| | | | | Male size × Female size | −0.1 | 0.6 | .901 |
| Homotypic mating or not | Socorro | 38 | 0.08 | Constant | 76.9 | 101.7 | .449 |
| | | | | Male size | −8.2 | 9.4 | .381 |
| | | | | Female size | −7.1 | 8.8 | .419 |
| | | | | Male size × Female size | 0.7 | 0.8 | .363 |

$R^2$ is the value of the Cox–Snell *R* value in the regression model. *B* is the Wald regression coefficient of each variable with its corresponding standard deviation (SD) and probability of the Wald test (*P*). $n_2$ is the sample size used for this analysis, which corresponds to the number of mating balls (experimental units) for which there was at least one mating and in which snail shell size was measured (as in Table 1; see text for further details).

**Table 3.** Estimates of assortative mating.

| | Soutoxuste | | | | | Portela | | | | | Socorro | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | $I_{PSI}$ | | | $G_{Isolation}$ | | $I_{PSI}$ | | | $G_{Isolation}$ | | $I_{PSI}$ | | | $G_{Isolation}$ | |
| | Estimate | SD | $n_1$ | *G* value | d.f. | Estimate | SD | $n_1$ | *G* value | d.f. | Estimate | SD | $n_1$ | *G* value | d.f. |
| All colours | −0.09 | 0.171 | 33 | 0.76 | 4 | −0.19* | 0.091 | 57 | 14.6** | 4 | −0.74*** | 0.157 | 38 | 18.2** | 4 |
| White | −0.06 | 0.177 | 33 | 0.1 | 1 | −0.32*** | 0.087 | 57 | 10.5** | 1 | −0.46** | 0.141 | 38 | 7.7** | 1 |
| Yellow | −0.07 | 0.185 | 33 | 0.2 | 1 | −0.01 | 0.099 | 57 | 0.1 | 1 | −0.37* | 0.168 | 38 | 3.5 | 1 |
| Brown | −0.04 | 0.134 | 33 | 0.1 | 1 | −0.08 | 0.092 | 57 | 0.8 | 1 | −0.56** | 0.122 | 38 | 9.4** | 1 |

Two complementary approaches are presented. First, the $I_{PSI}$ value is given both for the whole set of data ('All colours') and the data subdivided by pairs of colours (one given colour vs. each one of the remaining colours; 'White', 'Yellow', and 'Brown'). The probability of the estimate and the standard deviation (SD) are calculated by bootstrapping the original data (see Supporting Information, Table S3). Second, the maximum likelihood test ($G_{Isolation}$) compares the expected mating frequencies (assuming random mating from marginal mated frequencies) and observed matings by JMATING software (Carvajal-Rodriguez & Rolan-Alvarez, 2006). $n_1$ represents the total number of experimental units (mating balls) used in the experiment (as in Table 1). *$P$ < .05; **$P$ < .01; ***$P$ < .001.

The results in relation to the pattern of assortative mating (estimated through $I_{PSI}$), as well as their deviation from random mating (evaluated through G tests), are shown for the three localities in Table 3. All $I_{PSI}$ estimates were negative (inclined towards the negative assortative mating side), but they were statistically significant only for all colours and white vs. each of the rest of the colours in Portela and for all possible combinations in Socorro. G likelihood tests yielded very similar results to those obtained through bootstrapping estimates (except for the yellow colour in Socorro). Taken together, these show a general trend towards negative assortative mating in all three populations, even though only in two of them (Portela and Socorro) was this trend statistically significant.

For the two localities showing statistically significant negative assortative matings (Portela and Socorro), the distribution of matings was studied using PTI estimates (Rolán-Alvarez and Caballero 2000), which represent the maximum likelihood estimate of the success of a given mating type (see Table 4). PTI measures deviations from random mating in each of the nine possible mating types. In both localities, there is at least one diagonal cell (homotypic) that presents a value significantly lower than 1, which can be interpreted as evidence in favour of some kind of mate choice for negative assortative mating (i.e. homotypic matings having lower occurrence probability than that expected by chance). In Portela, this effect is seen in white (♀) vs. white (♂) matings, whereas in Socorro that would affect both yellow (♀) vs. yellow (♂) and brown (♀) vs. brown (♂). When the model comparison tool is applied, in both locations a preference for models that incorporate a coefficient of mate choice ($C_1$ and/or $C_2$) is selected. This suggests that choosing a model with only two *C* coefficients fits better to the data than a model considering all PTIs as well as the rest of the alternative models. In Portela, the preferred model incorporates two choice coefficients ($C_1$ and $C_2$), whereas in Socorro, despite showing a stronger choice trend, the preferred model shows only one choice coefficient affecting all diagonal matings ($C_1$), which could be interpreted as an a priori simple choice mechanism.

## DISCUSSION

Overall, this experiment provides additional support for the hypothesis that the negative assortative mating pattern for shell colour seen in wild populations of *L. saxatilis* is driven by a male mate choice mechanism for dissimilarity (Gefaell *et al.* 2021).

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**Table 4.** Analysis of the whole set of mating pairs using PTI (pair total isolation) estimates (calculated using JMATING), and the best mate choice model estimated by InfoMating for the localities with significant assortative mating (Portela and Socorro).

| | | Portela | | | | Socorro | | |
|---|---|---|---|---|---|---|---|---|
| | | **White (♂)** | **Yellow (♂)** | **Brown (♂)** | | **White (♂)** | **Yellow (♂)** | **Brown (♂)** |
| PTI | White (♀) | 0.39*** | 1.42 | 1.23 | White (♀) | 0.47 | 2.13 | 0.95 |
| | Yellow (♀) | 0.77 | 0.91 | 1.04 | Yellow (♀) | 0.71 | 0.24** | 0.95 |
| | Brown (♀) | 1.49 | 0.71 | 1.04 | Brown (♀) | 2.37 | 1.18 | 0*** |
| Model | White (♀) | $C_1$ | 1 | 1 | White (♀) | $C_1$ | 1 | 1 |
| | Yellow (♀) | $C_2$ | $C_1$ | 1 | Yellow (♀) | 1 | $C_1$ | 1 |
| | Brown (♀) | 1 | $C_2$ | $C_1$ | Brown (♀) | 1 | 1 | $C_1$ |
| Estimate | White (♀) | 0.6 | 1 | 1 | White (♀) | 0.17 | 1 | 1 |
| | Yellow (♀) | 0.57 | 0.6 | 1 | Yellow (♀) | 1 | 0.17 | 1 |
| | Brown (♀) | 1 | 0.57 | 0.6 | Brown (♀) | 1 | 1 | 0.17 |
| | | $C_1$ ± SD = 0.60 ± 0.171 | | | | $C_1$ ± SD = 0.17 ± 0.131 | | |
| | | $C_2$ ± SD = 0.57 ± 0.085 | | | | | | |
| | | Weight = 36% | | | | Weight = 30% | | |

The analyses were conducted on the corresponding summary tables shown in Supporting Information, Table S3. PTI includes the effects of sexual selection and sexual isolation in the population. Statistical significance is calculated by bootstrapping 10 000 times the frequencies of observed mating pairs. A model is chosen by InfoMating in each locality with their corresponding parameters ($C_1$ and $C_2$). The weight of the chosen model is also given for each locality. **$P < .01$; ***$P < .001$.

This, in turn, would support the negative frequency-dependent selection scenario for the maintenance of shell colour polymorphism in the wave-sheltered localities of the Ría de Vigo. We have found that, in the populations of two out of three of the studied localities (Portela and Socorro), a negative assortative mating pattern for shell colour like the one detected in the field is replicated under controlled laboratory conditions. The $I_{PSI}$ and PTI estimates support such a scenario, while the logistic model rules out the potential contribution of shell size to the obtained results. Taking into account the natural history and ecology of *L. saxatilis* (Reid 1996, Saltin *et al.* 2013), the experiment described in this study allows us to conclude that a male mate choice is responsible for the mating pattern observed. Furthermore, our model selection analysis shows that the best explanatory model for mate choice in two localities (Portela and Socorro) incorporates a choice coefficient ($C_1$) lower than that expected by random mating alone, affecting the whole diagonal of homotypic matings, as well as another coefficient lower than 1 ($C_2$) which affects only two types of matings in Portela. These coefficients mean that those particular mating combinations show a tendency to mate below the random expectation (which equals 1). This pattern of choice coefficients is only compatible with a negative assortative mating, as all homotypic (between identical colours) mating pair combinations show an occurrence probability < 1 (below random mating), thus backing the main conclusion obtained by the other analyses carried out.

However, despite the fact that we believe that the above conclusion—namely, that the negative assortative mating pattern detected is brought about by mate choice—is warranted, a few caveats also exist. The mating pattern observed for the snails of Soutoxuste, despite being slightly negative, cannot be distinguished from random mating (Table 3). This means that such a result departs not only from what is seen in the Portela and Socorro samples used for this experiment but also from what was detected in the field in such a locality. In fact, Gefaell *et al.* (2021) found a statistically significant negative assortative mating pattern for shell colour in the wild populations of Soutoxuste and Portela. Several potential explanations might be put forward to account for this inconsistency. One of these has to do with the effects of laboratory handling and storage of the organisms involved in the study. Discrepancies between laboratory and field results are a well-known phenomenon for ethologists and behavioural ecologists, with laboratory experiments usually entailing extreme manipulations that sometimes alter the behaviour of the studied organisms with respect to the one they display in the wild (Ruxton and Colegrave 2006, Bateson and Martin 2021). Moreover, even though the laboratory conditions are normally much more homogeneous than those of the field, there could still be a great deal of variation in at least some parameters, which in turn might affect specific subsets of organisms employed in an experiment. In our case, this could mean that certain extraneous variables might have affected the Soutoxuste snails more than those of the two other localities, even if we are not aware of any of these. *Littorina saxatilis* is extremely sensitive to environmental variation, particularly regarding its mating behaviour under artificial conditions. For instance, in a laboratory experiment with this species, Rolán-Alvarez *et al.* (2004) had to study mating attempts because the observed number of actual matings was too low to be representative. Although the experimental protocol for obtaining matings in the laboratory has been greatly improved since the introduction of the mating balls paradigm (Lau *et al.* 2021), this species and behaviour still constitute a troublesome case.

The second set of possible explanations for the observed discrepancies has to do with temporary starting differences in the natural populations of each locality. It could have been that the nutritional or parasitic status of the Soutoxuste snails at the time and year when sampled was worse than that of the remaining two populations, such that when they were taken to the laboratory, this influenced their mating behaviour. It is widely known that

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parasites influence the mating behaviour of their hosts (Moore 2002). Moreover, there is evidence that infection by trematodes alters the sexual and nonsexual behaviour of at least another species of the genus *Littorina* (*L. littorea*; Saur 1990, Erlandsson and Kostylev 1995, Davies and Knowles 2001, Seaman and Briffa 2015). For instance, Erlandsson and Kostylev (1995) observed that male mucus trail following was greatly reduced in a male *L. littorea* following a highly infected female. It cannot be ruled out that this might have affected the behaviour of the Soutoxuste snails in this experiment as well. The fact that the total number of mating balls in which there was at least one mating was lower in Soutoxuste than in the mean of all three localities ($M$ = 60.8%, SD = 3.5 in Soutoxuste vs. $M$ = 67%, SD = 6.8 in the three populations; see Supporting Information, Table S4) supports this possibility. However, both this and the former explanations are mere guesses for which there is no conclusive evidence and that should be studied in the future.

Finally, a third potential explanation could be that the fitness advantage of being a rare colour in Soutoxuste might not be as great compared to the other two localities (Portela and Socorro), perhaps because shell colours in Soutoxuste may also have an impact on fitness in other ways in this specific population of *L. saxatilis*. Although we neither can rule out this possibility, currently there is no direct data in support of such a hypothesis. Quite the contrary, this contradicts the results obtained in the field study carried out in Soutoxuste and Portela, which found negative assortative mating in both localities as well (Gefaell *et al.* 2021). Furthermore, all known ecological variables in Soutoxuste, as well as all the relevant morphological traits of the snails of that locality, are virtually identical to those of Portela and Socorro, thereby making it difficult to come up with plausible scenarios regarding the potential factors that could be exerting specific fitness effects on the snails of Soutoxuste.

The fact that there is an incongruence in the results obtained for one out of three localities suggests that the best way to settle this issue would be to conduct a mate choice experiment in the field. That would remove any possible extraneous laboratory factors from affecting the behaviour of the organisms while providing more ecological validity to the results obtained (Bateson and Martin 2021). If the same overall pattern as the one observed both in Gefaell *et al.* (2021) and in this study persists, then we would expect a corroboration of male mate-driven negative assortative mating for shell colour.

Even in the face of the limitations of our laboratory experiment, we nonetheless provide evidence that the colour polymorphism seen in the wave-sheltered localities of the Ría de Vigo is probably maintained through negative frequency-dependent selection driven by mate choice for dissimilarity. This raises the question of how the colour of females is detected by male *L. saxatilis* individuals. So far, all evidence suggests that snails of the genus *Littorina* (including *L. saxatilis*) do not have colour vision (Gefaell *et al.* 2021), which means that, if there is any mate choice for shell colour, there should be an indirect way for male snails to detect and select those females with a dissimilar coloration. It has been posited that there might be differences in the metabolic composition of the mucus trails between the colour morphs of *Littorina* snails (Estévez *et al.* 2020), but this hypothesis has not been tested so far. In addition to this problem, the existence of mate choice for dissimilar shell colours also prompts the question of what fitness benefits derive from shell colour. As we have argued, shell coloration might contribute to fitness by providing important information to either conspecifics and/or predators (communication), better camouflage (crypsis), or a more efficient control of body temperature (thermoregulation). Apparently, there are no a priori obvious fitness differences between the colour morphs of *L. saxatilis* from the wave-sheltered localities of the Ría de Vigo when it comes to crypsis or any selective pressure directly related to shell colour per se (E. Rolán-Alvarez, personal observation). This opens the door to the possibility of shell colour just being a trait correlated with other fitness-enhancing traits (Gefaell *et al.* 2023a). In that respect, a previous study on *L. saxatilis* from Sweden (Koch *et al.* 2021) was able to associate one of the colours present in these populations to a specific chromosomal inversion, which was also linked to other adaptive shell traits. Therefore, a plausible hypothesis could be that the true target of (sexual) selection might be a trait genetically linked to shell colour that the snails somehow manage to detect, but not shell colour itself. For this reason, future studies should not only be aimed at settling whether a male mate choice-driven negative assortative mating for shell colour really exists in these populations of *L. saxatilis* but also at studying its potential genetic underpinnings.

## SUPPLEMENTARY DATA

Supplementary data are available at *Biological Journal of the Linnean Society* Journal online.

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## CONFLICT OF INTEREST

The authors declare that they have no conflict of interest.

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## DATA AVAILABILITY

Raw data are available in the Figshare repository: https: //doi.org/10.6084/m9.figshare.23515020.v1.

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